

Observational studies

An observational study records conditions without assigning treatments.

Students choose whether to attend tutoring. We compare their exam scores. What kind of study is this?

Experiments

An experiment assigns treatments and measures outcomes.

**A teacher randomly assigns two practice formats.
What makes this an experiment?**

Experimental units

An experimental unit is an individual or object that receives a treatment.

Thirty printers are each assigned one of two paper types. What are the experimental units?

Explanatory variables

The explanatory variable is the proposed influence on an outcome.

A study compares two paper types for their effect on printer jams. What is the explanatory variable?

Response variables

The response variable is the measured outcome.

How could we measure printer performance in the paper-type study?

Treatments

Treatments are the experimental conditions assigned to units.

A printer study uses standard, recycled, and coated paper. What are the treatments?

Lurking variables

A lurking variable is an unaccounted-for influence on a study.

**Students who choose tutoring earn higher scores.
What other influence might explain the difference?**

Confounding

Confounding prevents us from separating the effects of different influences.

One practice format is taught only by Teacher A, the other only by Teacher B. What is confounded?

Random assignment

Random assignment uses chance to allocate treatments.

How could we assign 40 volunteers to two practice formats?

Chance imbalance

Forty volunteers are randomly assigned to two practice formats.

Must the two groups begin with exactly equal prior knowledge?

Random sampling or assignment?

A researcher randomly selects 100 students, then lets each choose a practice format.

Which part is random sampling, and is there random assignment?

Control groups

A control group provides a comparison condition.

What would be a useful control group for a study of a new practice app?

Placebos

A placebo resembles a treatment but lacks its active component.

Why might an inactive version of a treatment be useful in a comparison?

Placebo effects

A placebo effect is a response associated with expectations of treatment.

How could a participant's expectations change a self-reported outcome?

Blinding

Blinding withholds treatment identities from specified people.

Why might a grader receive assignments without knowing each student's practice group?

Double blinding

In a double-blind study, participants and relevant researchers do not know treatment assignments.

Could students be blinded to whether their practice uses paper or an app?

Replication

Replication applies each treatment to multiple independent experimental units.

Why would one printer per paper type be weak evidence?

Causal conclusions

Students who voluntarily use a new study app score higher than students who do not.

Does this establish that the app caused the difference?

A study design

Forty volunteers will compare two versions of a practice app using the same final quiz.

How would you design a fair experiment?

Participation and consent

A researcher tells students that refusing a study will lower their course grades.

What is ethically wrong with this recruitment?

Privacy

A public spreadsheet lists participant names beside individual quiz scores.

How should these data be handled before sharing?

Honest reporting

A researcher deletes accurate low scores because they weaken the desired conclusion.

What is wrong with this decision?

Selective reporting

A team runs ten comparisons and reports only the one with its preferred result.

What information does the audience need to judge the evidence?